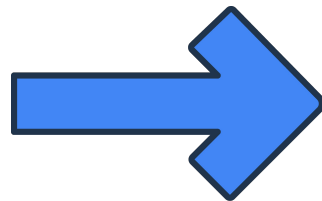


Embeddings

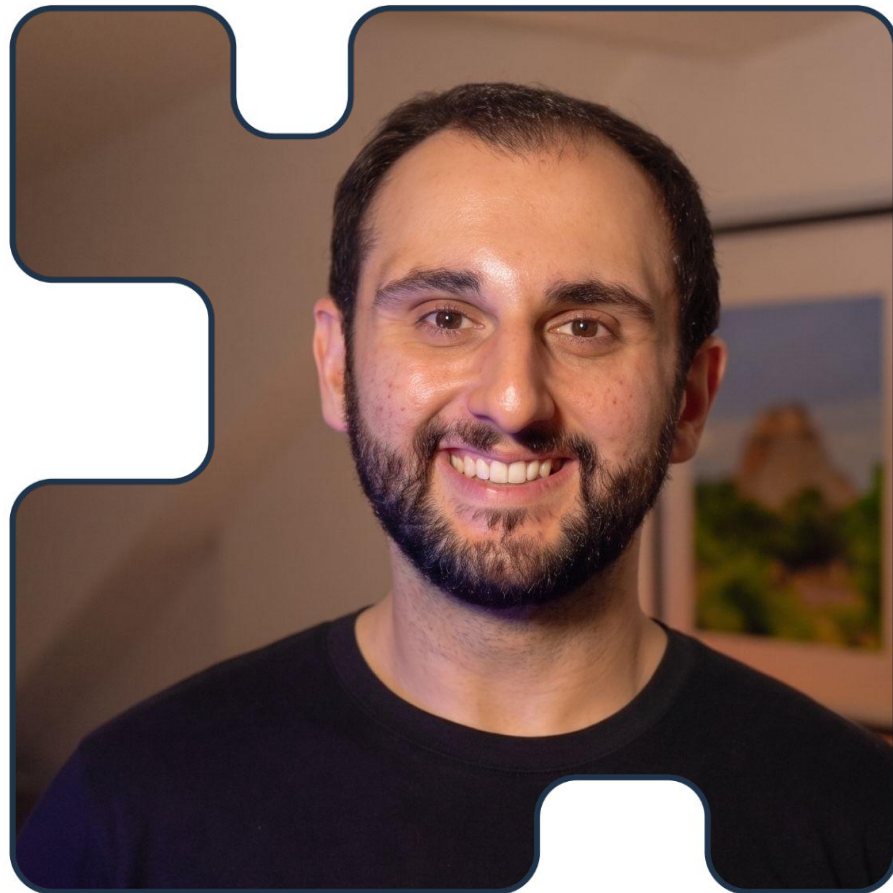
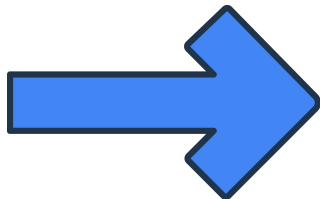
How to make search actually understand you





Alexander Amin

Google Developer Expert
Cloud & Cloud AI



What is **vector search**?

Consumer use cases

- ✓ Find a product from a vague picture.
- ✓ Identify a song with a piece of lyric.
- ✓ Seek vacation recommendations.



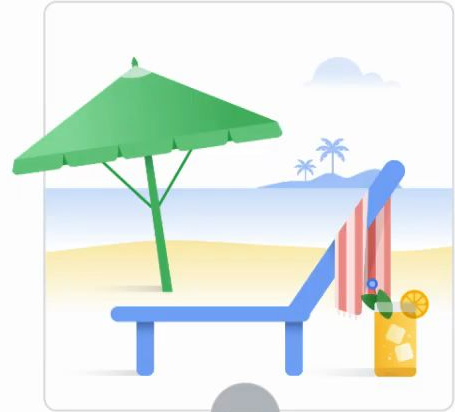
Vector search:
Semantic similarity

Keyword search

Keyword search

- Matches keywords effectively.
- Does not understand word meanings.
- Lacks knowledge of context.

summer tops



Semantic search

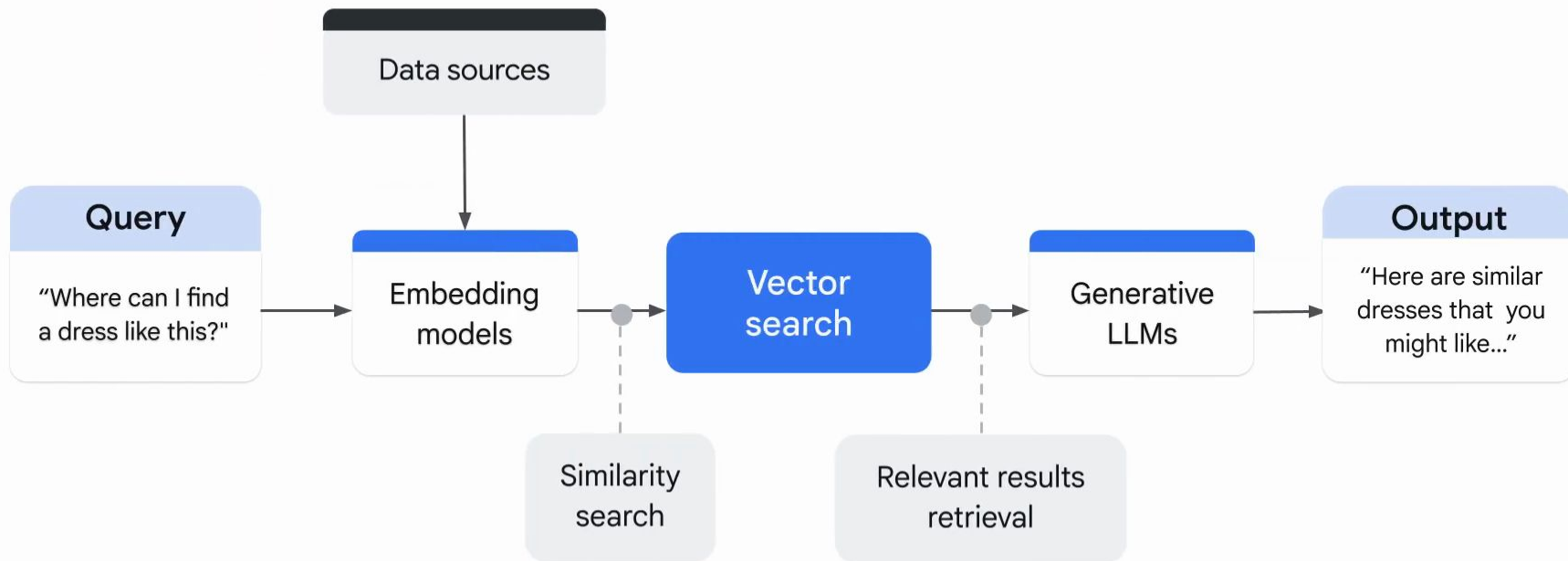
Keyword search

- Matches keywords effectively.
- Does not understand word meanings.
- Lacks knowledge of context.

Semantic search

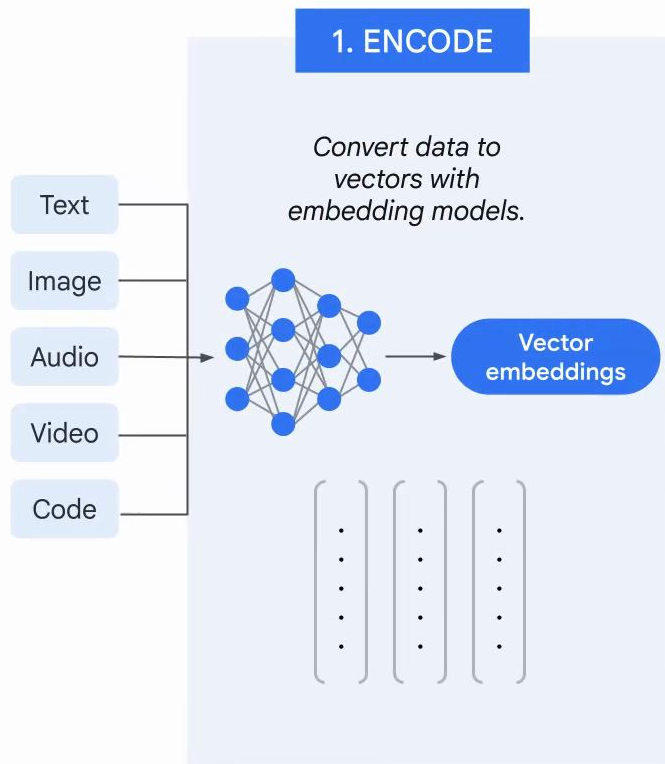
- Finds semantically similar items.
- Transforms data to meaningful “embeddings”.
- Understands context.

Vector search is critical in AI applications



How does **vector search** work?

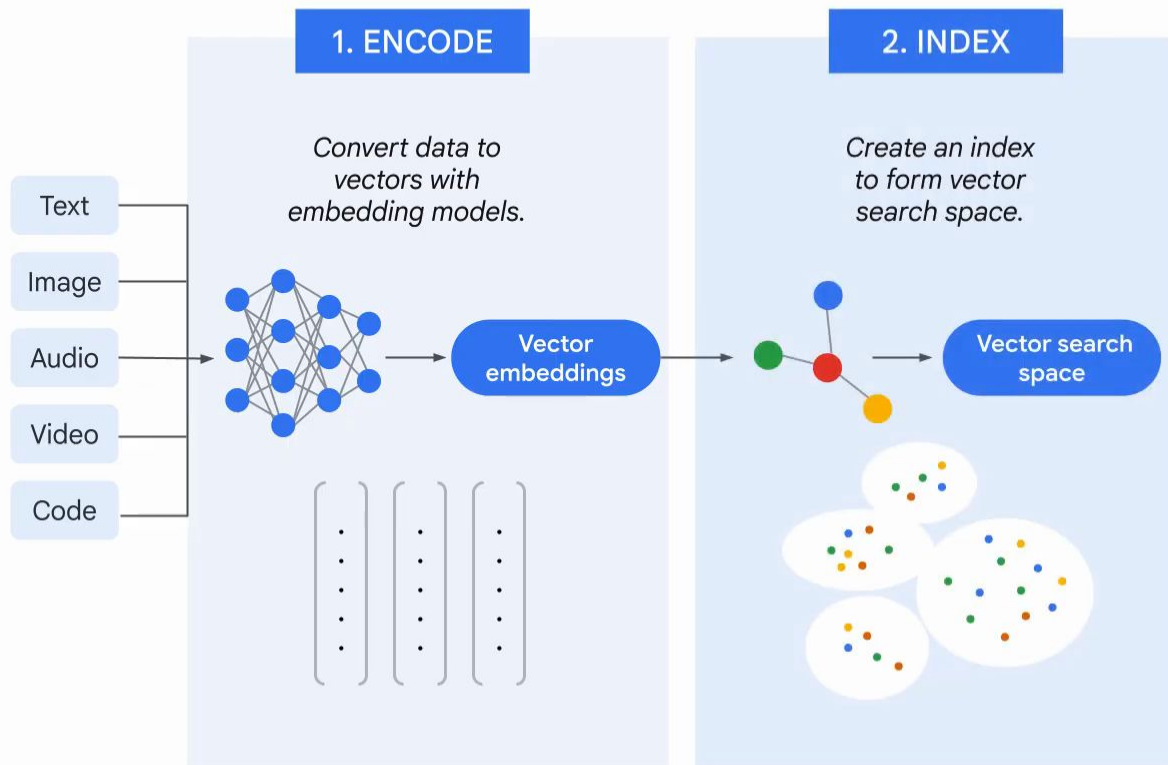
Vector search process



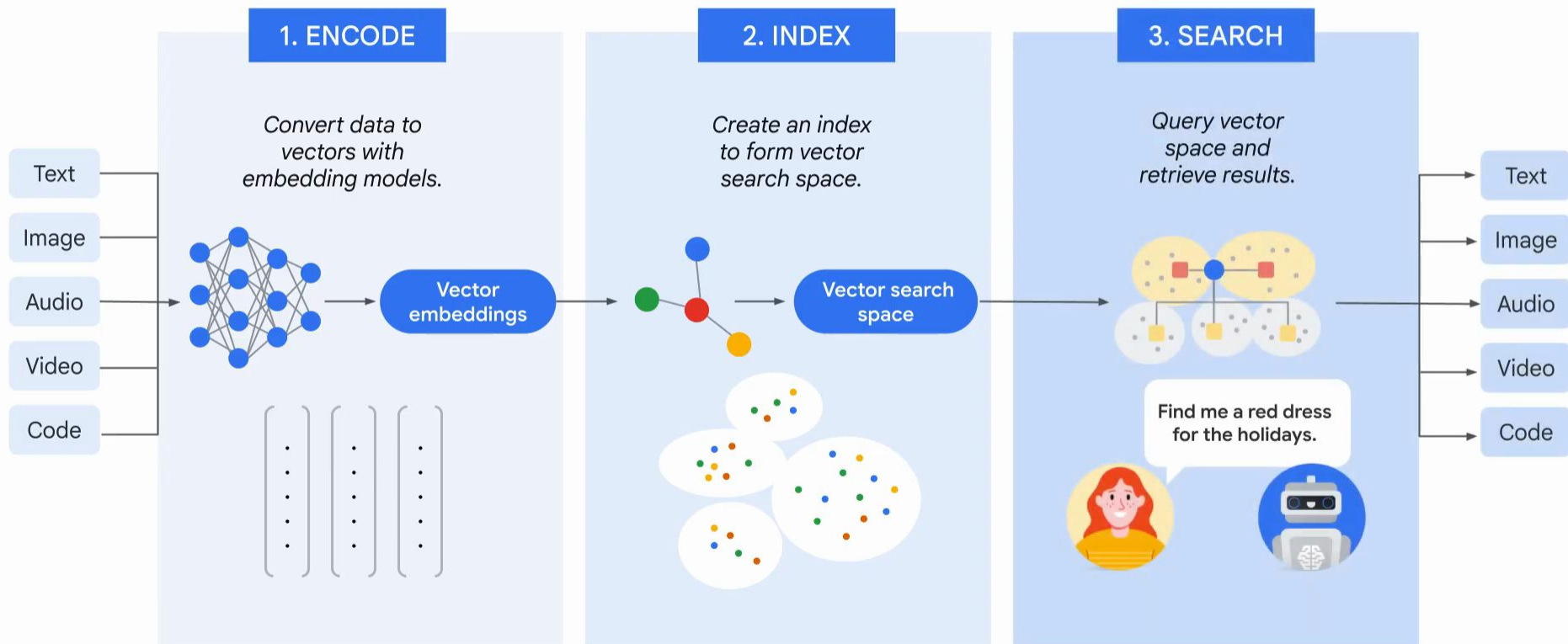
Embedding Generated by gemini-embedding-001 (Size 3072)

[0.021900404,0.037497032,0.0010875815,-0.06999241,0.025528956,0.014025146,0.009787402,0.021767298,0.034139913,-0.028766846,-0.015751697,0.0032291168,-0.020500036,0.0150761325,0.106516324,0.022041466,-4.7055135e-05,0.003252801,-0.017746095,-0.0029796567,0.0031421203,-0.027407313,-0.0134469755,0.015858816,0.014293725,-0.00840819,0.021698195,0.0130695,0.017240113,-0.008730424,-0.0001350386,0.0037008512,0.013249227,0.010248512,0.0003332556,0.0038460747,-0.00017510817,0.009161036,-0.009318005,0.0077442327,-0.024243366,-0.008619575,-0.005152853,0.0043334463,0.008662574,0.005597216,0.007937557,-0.021392189,-0.013265949,0.022947634,-0.0027800319,-0.013015706,-0.014612949,-0.1742421,-0.0065960353,0.0042746644,-0.025696507,0.039846126,0.0124945035,1.635169e-05,-0.02209765,0.042150706,-0.0075065973,-0.030125637,0.018377297,0.006796165,-0.0074976226,0.024125831,-0.019929273,-0.027853934,0.0026122232,-0.01183689,0.029348532,-0.024136772,-0.023128763,-0.0034410998,-0.00824666,-0.008411613,0.0099716,-0.02543757,0.008940514,-0.003819695,-0.002431808,-0.014856651,0.011849413,-0.001757723,-0.020012315,-0.010513511,0.01094601,-0.0004288963,-0.00028937357,0.005351356,0.023702674,-0.03661723,0.010781953,0.019263826,0.0033291362,-0.004267097,-0.0032952165,-0.006607751,-0.0077254465,-0.005798446,0.017218897,-0.026613137,0.039661374,-0.019865291,0.008387136,0.019626807,-0.02412999,0.005639717,-0.036912333,-0.032330863,0.035233118,-0.016932791,-0.00774114,-0.0025351024,-0.014813104,-0.033173658,0.013331434,-0.0031507679,0.0032035476,-0.013391962,-0.016575703,0.018118236,-0.013607977,0.005245942,0.022080794,0.014048013,0.028970053,-0.0111869415,0.011500652,-0.02046716,-0.009297797,-0.011082396,0.01099034,0.032973032,-0.0031819667,-0.019892491,0.017049765,-0.0708842,0.01731937,0.012278637,0.0123554235,0.017133206,-0.027939403,0.018629225,0.038838953,-0.0040503894,0.016427094,0.0005230735,0.0028389096,0.023750572,0.005046381,0.0062940726,0.008560822,0.012893724,-0.0044051735,-0.003143098,-0.03566533,-0.012432426,-0.013023975,-0.01798856,-0.004885465,-0.0027495616,-0.0009863942,0.007455733,0.03504787,-0.023396067,0.0004420177,...]

Vector search process



Vector search process



Two key technologies

01

How to encode data?

How to convert multimodal data to a representation that captures semantic meanings?

Embeddings

02

How to index and search data?

How to build a search space that enables fast and efficient search?

Vector search

How can you represent text in a
numeric format while **retaining its meaning**?



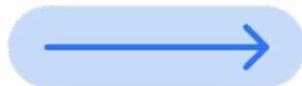
Text

Text embeddings



Vectors that contain meaning

How do you describe a dog?



Breeds

...

...

...

Physical stats

...

...

Hair

...

Relationships

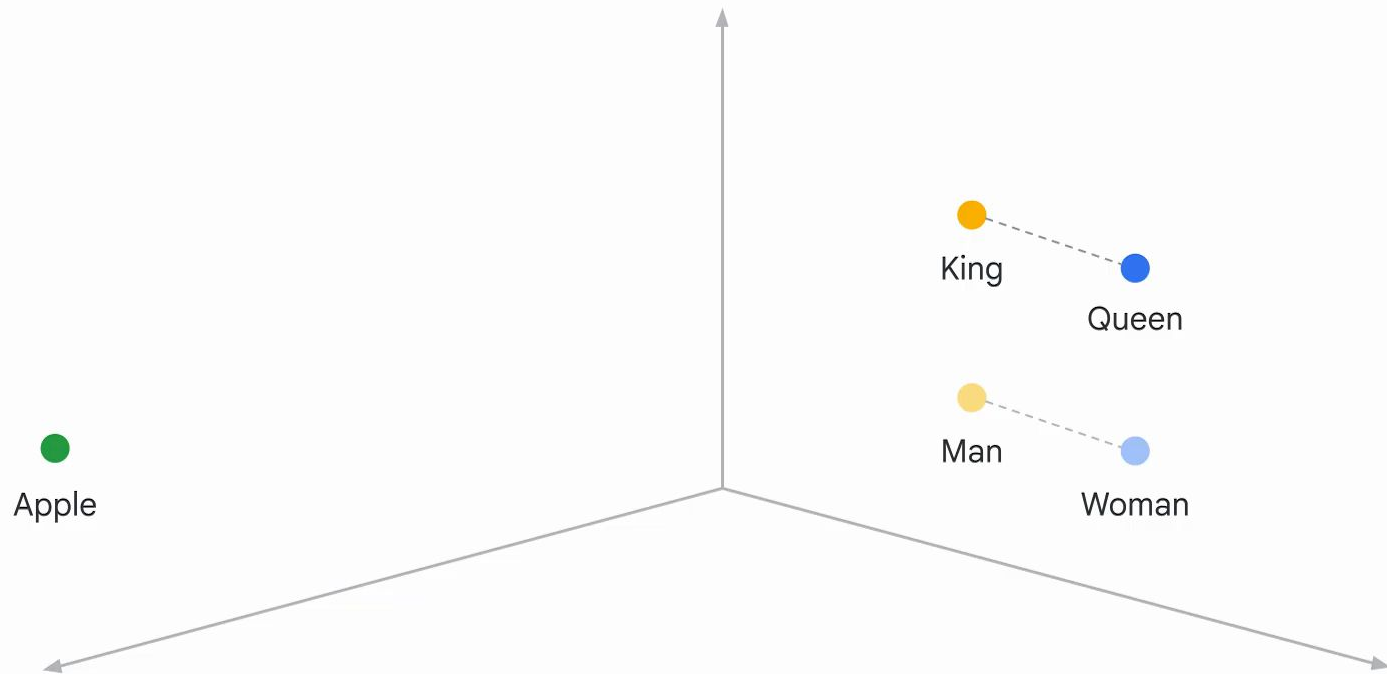
...

...

...

Behaviors

...



$$\text{King} - \text{Man} + \text{Woman} = \text{Queen}$$

Text embedding

A technique to encode text
into meaningful vectors.

Benefits of text embeddings



They contain dense representation.



Low dimension



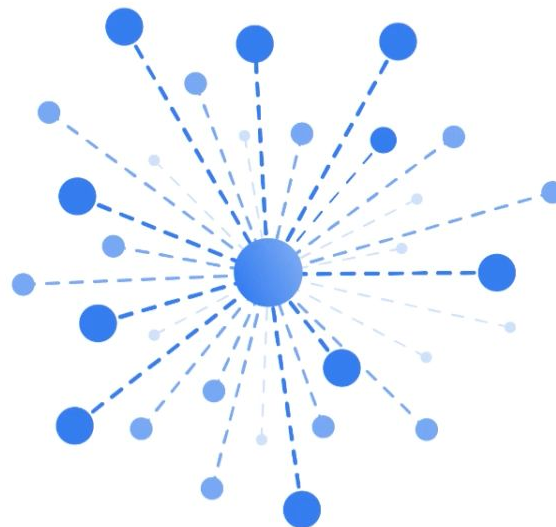
Dense vectors



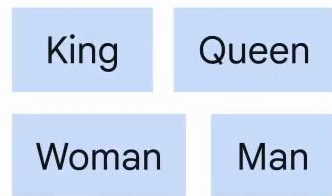
They capture the semantic meaning of words.



Distributed representation



Dense embeddings



Input (text)

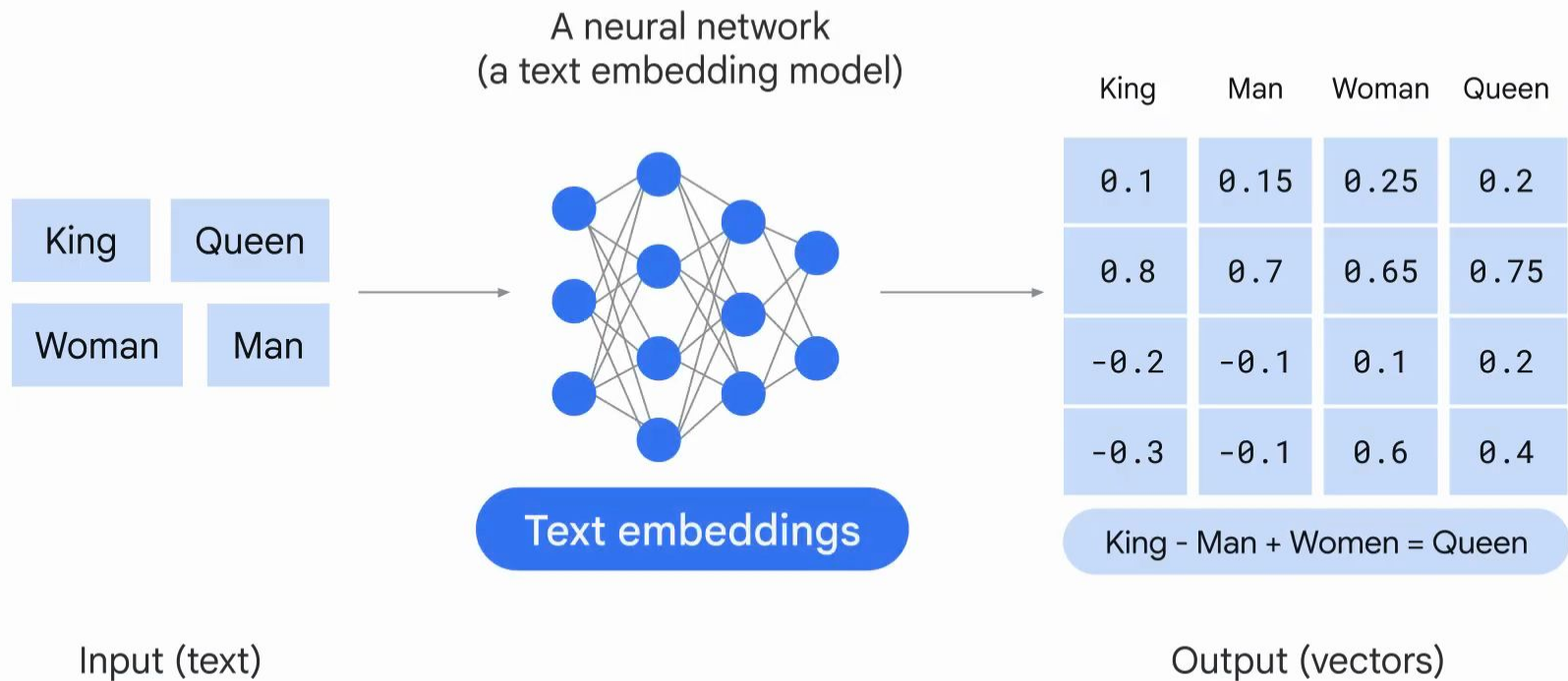


Diagram showing the output vectors for the input text. An arrow points from the puzzle piece to a table of numerical values.

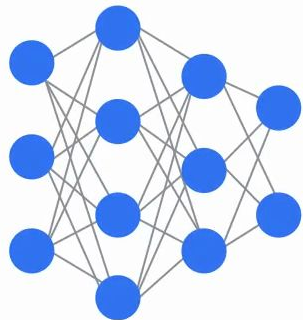
King	Man	Woman	Queen
0.1	0.15	0.25	0.2
0.8	0.7	0.65	0.75
-0.2	-0.1	0.1	0.2
-0.3	-0.1	0.6	0.4

King - Man + Woman = Queen

Output (vectors)



A neural network
(a text embedding model)



Text embeddings



word2vec

by Google



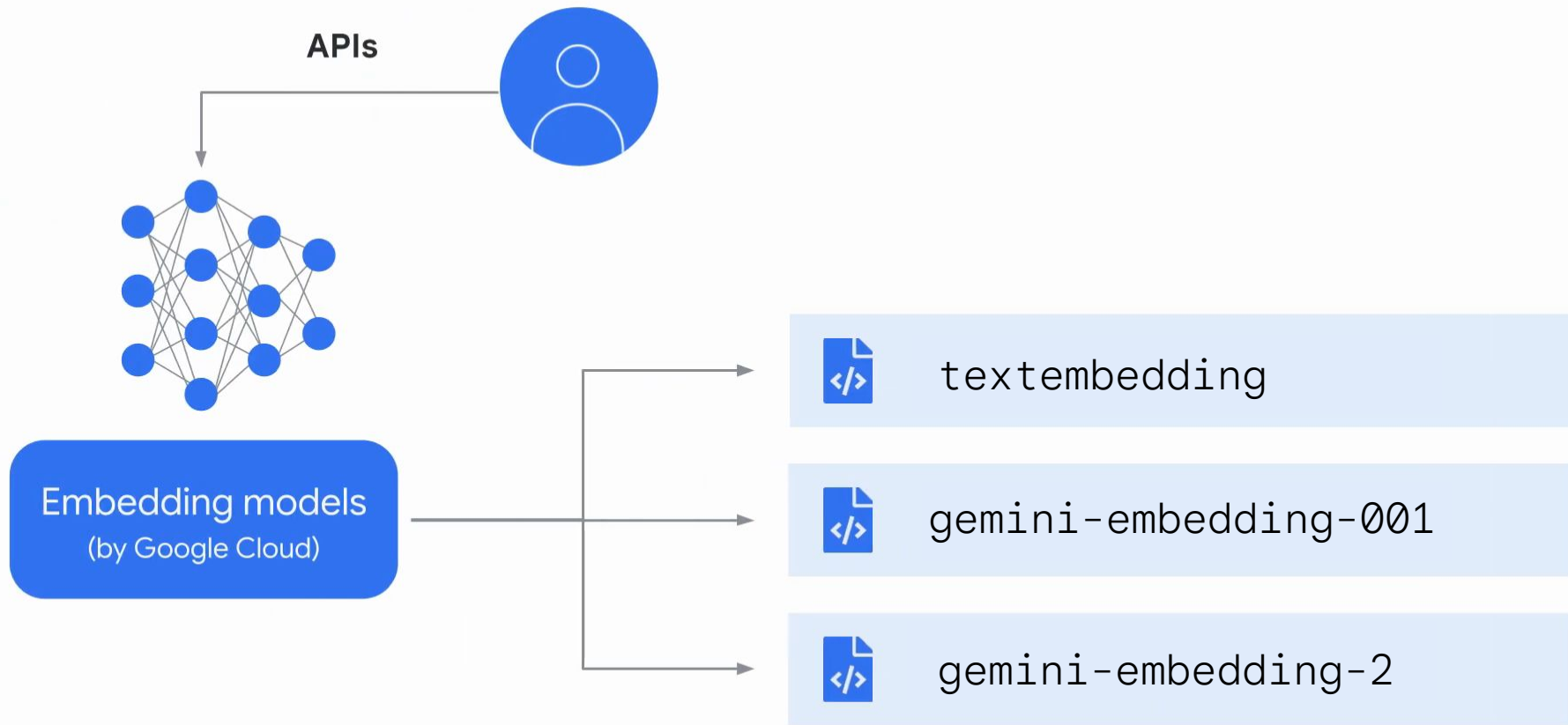
GloVe

by Stanford



fastText

by Meta



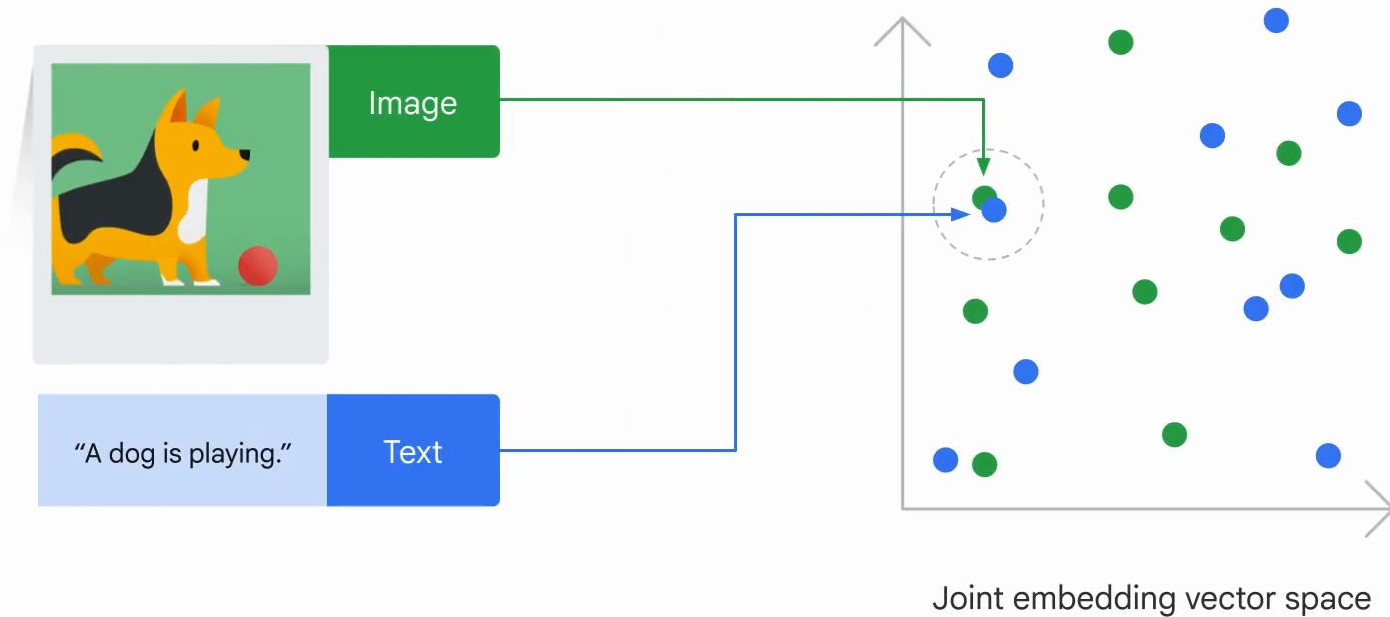
```
from google import genai

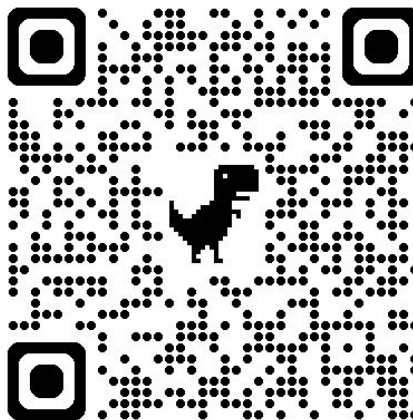
client = genai.Client()

result = client.models.embed_content(
    model="gemini-embedding-2",
    contents="What is the meaning of life?"
)

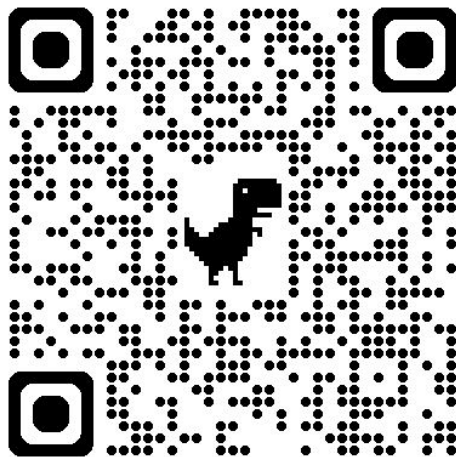
print(result.embeddings)
```

Multimodal embeddings





https://www.skills.google/course_templates/939/labs/637179



<https://alexamin.de/embeddings.ipynb>